

Aarya Pant

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PROFILE

Geomatics Engineering student with a strong foundation in GIS, remote sensing, and web development. Interested in combining GIS, UAV, AI to solve environmental issues and infra planning. Committed to utilizing the most current geospatial technology to further community-based initiatives. Eager to apply technical expertise in spatial analysis and data visualization to solve real-world challenges.

SKILLS

- GIS
- Remote Sensing
- Photogrammetry
- Surveying and Geodetic Science
- Web GIS
- Web Development
- Database
- HecRAS
- Python

EDUCATION

Kathmandu University | *BE in Geomatics Engineering*

Dhulikhel | **Nov 2021 – Present**

- Semester-wise GPA:

1st Sem: 2.88

4th Sem: 3.88

7th Sem: 3.96

2nd Sem: 3.36

5th Sem: 4.00

3rd Sem: 3.85

6th Sem: 3.85

NASA Higher Secondary School | *High School*

Tinkune | **April 2019 – May 2021**

- Cumulative GPA: 3.46

Jaycees Everest English School | *Schooling*

Dhangadhi | **2009 – 2019**

- Cumulative GPA: 3.45

EXPERIENCE

Student Participant | **Australia-Nepal Mobility Program**

Aug 15 – Aug 25

- Participated in a month-long academic and cultural exchange program with the University of Southern Queensland (UniSQ) under the New Colombo Plan (NCP). Conducted joint fieldwork with UniSQ and LMTC, including instrument calibration, levelling, topographic and cadastral surveys, data processing, and drone applications.

Treasurer | **Geomatics Engineering Society**

Sept 2024 - Present

- Managed society finances with transparency, prepared budgets, and supported event organization and coordination.

Trainer | **Geowomen: Promoting Spatial Data Literacy**

Feb 2025

- Led a two-day virtual training session organized by NAXA, Robotics Association of Nepal (RAN), and Women in Geospatial Nepal (WiG Nepal).
- Facilitated hands-on mapping activities using OpenStreetMap (OSM) and the OSM Tasking Manager.
- Guided participants in mapping buildings and roads to enhance rural and urban accessibility.

- Introduced advanced remote mapping techniques and mobility assessments to address local challenges.

Sudurpaschim Samaj (SuPa Samaj, KU) | Media and Marketing Head

Feb 2024 - Present

- Led media, publicity, and outreach initiatives, managed social platforms, and promoted cultural and academic events to strengthen community engagement.

G17 Global | G17 University Ambassador for SDG 5 (Gender Equality)

June 2024 – June 2025

- Advocated for gender equality and women's empowerment through awareness, campaigns, and student engagement initiatives at the university level.

PROJECTS

- **Vulnerability and Risk Zonation Dashboard** [docs](#)
Developed a mini geoportal integrating hazard, exposure, and vulnerability datasets for interactive flood and disaster risk visualization using open-source web technologies.
- **AQI Monitoring Dashboard**
Designed a real-time air quality monitoring dashboard with interactive charts to visualize pollutant levels, AQI trends, and weather parameters for data-driven environmental insights.
- **Windfarm Site Selection Suitability using AHP: Myagdi District** [docs](#)
Conducted multi-criteria site selection using AHP, integrating wind, slope, land use, and accessibility data to identify optimal wind energy sites.
- **Shortest Route Analysis using PgRouting: Kathmandu to Dhangadhi**
Performed network analysis using PgRouting to determine the most efficient routes based on road networks and distances.
- **Konvertor - Coordinate Systems Tool**
Built a tool for coordinate conversion and system transformation to improve geospatial data interoperability and accuracy in JAVA.
- **Flood Risk Assessment of Bagmati River**
Analysed flood-prone zones using GIS and hydrological data, producing risk maps for disaster planning and mitigation.
- **Impact of Land Fragmentation on Agriculture**
Studied how land fragmentation affects agricultural productivity in Nepal, analyzing field and secondary data to suggest sustainable management strategies.
- **NDVI Analysis of Dhulikhel**
Applied remote sensing to assess vegetation health and land cover changes, generating NDVI maps for monitoring ecological trends.
- **Open Mapping Hub Asia-Pacific Field Participant: Urban Rooftop Farming**
Conducted fieldwork in Banepa Municipality, collecting geospatial data and supporting sustainability assessments for urban rooftop farming.

TECHNOLOGIES

- Languages: C++, C, SQL, JavaScript, Python, Java
- Technologies: QGIS, Envi, PiX4D, HEC-RAS, HEC-HMS, VS Code, Microsoft Office Suite, Flask, Torch, NumPy, Parcel Editor

SOFT SKILLS

- Communication: Public speaking, Negotiation, Active listening.
- Leadership Teamwork: Decision-making, Conflict resolution, Collaboration.
- Problem-Solving: Analytical thinking, Creativity, Troubleshooting.
- Time Management: Prioritization, Multitasking, Meeting deadlines.
- Adaptability: Handling pressure, Flexibility, Crisis management